

Unconfoundedness II: Propensity Scores and Regression

BUSS975: Causal Inference in Financial Research

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Outline

1. Propensity scores
2. Regression
3. What matching cannot do
4. The NSW demonstration

1. Propensity Scores

Where Lecture 3a left us

- Unconfoundedness says: **compare treated and controls with the same X .**
- **Subclassification** and **matching** suffer from the curse of dimensionality.
- Must we match on all of X ? The assumption is stated conditional on X , so it seems so.
- **Rosenbaum and Rubin (1983): no.** One scalar computed from X does the whole job.

Defining the propensity score

- **Propensity score:** $e(X) = \Pr(D = 1 \mid X)$
 - ▶ “the conditional probability of assignment to a particular treatment given a vector of observed covariates” (Rosenbaum and Rubin 1983)
- Three things it is:
 - ▶ A **probability**, so it lives in $[0, 1]$.
 - ▶ A **conditional** probability — read it as: *of the units that look like this on X , what fraction are treated?*
 - ▶ A description of the **assignment mechanism**, not of the outcome.
- Two results follow: the **balancing property** (conditional on $e(X)$, treated and control look alike on X) and the **propensity score theorem** (conditioning on $e(X)$ does what conditioning on all of X does).

The balancing property

Claim. $D \perp\!\!\!\perp X \mid e(X)$

D is binary, so its distribution is summarised by its mean; the claim is equivalent to $E[D \mid X, e(X)] = E[D \mid e(X)]$. We show **both sides equal** $e(X)$, and hence each other.

Left side. $e(X)$ is a function of X , so conditioning on it adds nothing to X :

$$E[D \mid X, e(X)] = E[D \mid X] = e(X) \quad (\text{definition of } e(X)).$$

Right side. Apply the **law of iterated expectations**, $E[A \mid C] = E_B[E[A \mid B, C] \mid C]$, with $A = D$, $B = X$, $C = e(X)$:

$$E[D \mid e(X)] = E_X \left[\underbrace{E[D \mid X, e(X)]}_{= e(X) \text{ by the left side}} \mid e(X) \right] = E_X[e(X) \mid e(X)] = e(X). \blacksquare$$

The mean version, $E[X \mid D=1, e(X)] = E[X \mid D=0, e(X)]$, is what a balance table tests.

The propensity score theorem

Claim. If $(Y^1, Y^0) \perp\!\!\!\perp D \mid X$, then $(Y^1, Y^0) \perp\!\!\!\perp D \mid e(X)$

In mean form, $\underbrace{E[D \mid Y^1, Y^0, X] = E[D \mid X]}_{\text{assumed}} \implies \underbrace{E[D \mid Y^1, Y^0, e(X)] = E[D \mid e(X)]}_{\text{to show}}.$

The right-hand side is $e(X)$ by the previous slide, so only the left-hand side is left to prove:

$$\begin{aligned} E[D \mid Y^1, Y^0, e(X)] &= E_X \left[E[D \mid Y^1, Y^0, X, e(X)] \mid Y^1, Y^0, e(X) \right] && \text{law of iterated exp.} \\ &= E_X \left[E[D \mid Y^1, Y^0, X] \mid Y^1, Y^0, e(X) \right] && e(X) \text{ is a function of } X \\ &= E_X \left[\underbrace{E[D \mid X]}_{= e(X)} \mid Y^1, Y^0, e(X) \right] = e(X). \blacksquare && \text{unconfoundedness} \end{aligned}$$

/normalize

The *ATE* is identified by comparing treated and control units at the same **score** rather than the same X :

$$ATE = E \left[E[Y \mid D=1, e(X)] - E[Y \mid D=0, e(X)] \right]$$

Estimating the score

- You know the **true** score only if you ran the experiment. Otherwise estimate it: $\Pr(D_i=1 \mid X_i) = F(\beta X_i)$, by **logit or probit** — not OLS, which extrapolates linearly and will hand you “probabilities” outside $[0, 1]$.
- Judge the specification by **balance achieved**, not by fit. A score with a high pseudo- R^2 but poor balance is a bad score; the reverse is fine.
- Fit, check balance, add interactions and polynomials where balance fails, refit. A legitimate specification search: it never looks at Y , so the stage is **blind to the outcome**. Say so in the paper.
- **The score creates no support that X lacks.** If a cell is all-treated or all-control, collapsing X to a scalar hides that; it does not fix it.

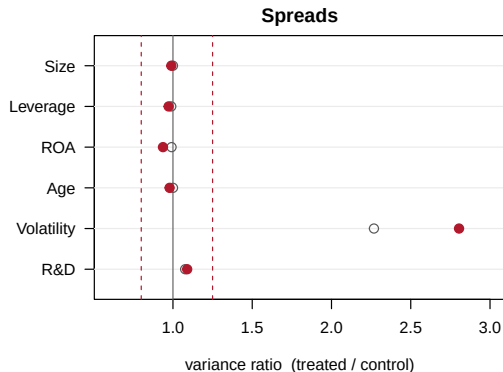
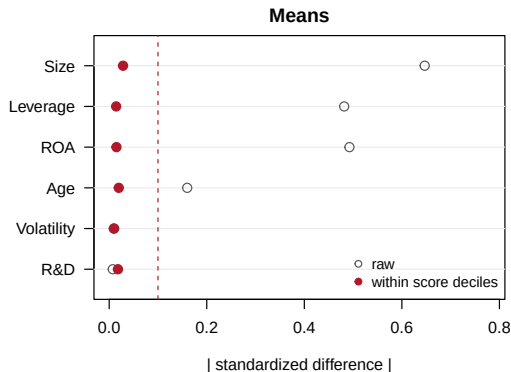
Checking balance: not a t -test

The balancing property is checkable. Report it covariate by covariate, before and after.

$$\text{SMD} = \frac{\bar{X}_1 - \bar{X}_0}{\sqrt{(s_1^2 + s_0^2)/2}}, \quad \text{VR} = \frac{s_1^2}{s_0^2}$$

- **Standardized mean difference.** $|\text{SMD}| < 0.1$ is good.
 - ▶ Keep the denominator **fixed at its raw-sample value** when recomputing after weighting, or the yardstick moves with the thing you are measuring.
- **Variance ratio.** Means can match while spreads do not. Rubin: stay inside $[0.5, 2]$, ideally $[0.8, 1.25]$.
- **Do not use t -tests.** Balance is a property of *this sample*, not a population hypothesis — and p confounds imbalance with sample size (Imai, King and Stuart 2008).

A balance plot: means are not enough



Selection here is linear in Size, Leverage, ROA, Age — and **quadratic in Volatility**. The score was fitted with linear terms only.

Reading the balance plot

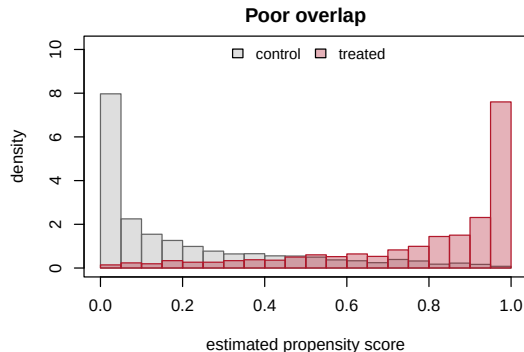
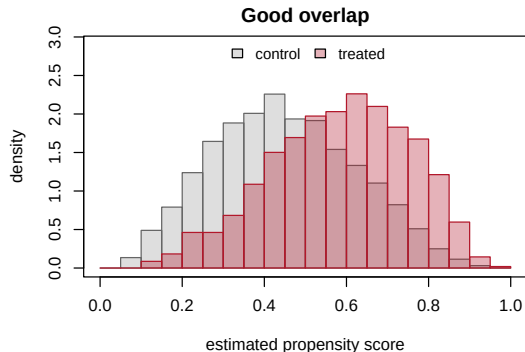
- **Left panel, the success.** Four covariates start badly imbalanced — Size at 0.65, Leverage and ROA near 0.49. Compared *within deciles of the score* all fall under 0.03. This is the plot referees want, and where most papers stop.
- **Left panel, the trap.** Volatility looks *perfect* from the start: $SMD = -0.010$, and a t -test returns $p = 0.72$. On the evidence of means, nothing to see.
- **Right panel.** Its variance ratio is **2.27** — treated firms have more than twice the spread. Extreme volatility drives selection in *both* tails.
- **Conditioning on the score makes it worse**, $2.27 \rightarrow 2.80$, because the fitted score never contained vol^2 . Add that one term and the ratio falls to **1.07**.
- Judge the score by **balance achieved, not by fit** — and balance means the whole distribution. Check squares and interactions, not just levels.

Three things to do with the score

Once every unit has a $\hat{e}(X_i)$, three standard uses:

1. **Diagnose.** Plot $\hat{e}$ separately for treated and control units. This is the common-support check, and it costs nothing.
2. **Weight.** Reweight the sample to build a pseudo-experiment — inverse probability weighting.
3. **Match.** Pair treated to control units with similar scores, then difference — propensity score matching.

Visualizing overlap



Right panel: a treated mass at $\hat{e} \approx 1$ with no controls, and a control mass at $\hat{e} \approx 0$ with no treated. Those units have **no counterfactual**. Any estimate covering them is extrapolation, not comparison.

This plot belongs in every such paper

- Common support is verifiable, unlike unconfoundedness.
- What to do when it looks like the right panel:
 - ▶ **Trim** extreme scores, or impose a caliper: drop units with $\hat{e} \notin [0.1, 0.9]$. A rule of thumb, but a defensible one (Crump, Hotz, Imbens and Mitnik, 2009).
 - ▶ **Report the change in estimand.** After trimming you describe a subpopulation. Say how many observations went and how the treated sample shifted.
 - ▶ Consider retreating to the *ATT*, which only needs overlap in one direction.
- What *not* to do: report the estimate and stay silent. Regression will extrapolate into a region with no data and give a standard error reflecting none of it.

Inverse probability weighting

Create a balanced pseudo-population by inverse probability weighting. Write $\hat{e}_i$ for $\hat{e}(X_i)$ and $\pi = \Pr(D=1)$.

$$ATE = E[Y_1 - Y_0] = E\left[\frac{D Y}{e(X)}\right] - E\left[\frac{(1-D) Y}{1-e(X)}\right]$$

$$ATT = E[Y_1 - Y_0 | D = 1] = \frac{1}{\pi} \left(E[DY] - E\left[\frac{e(X)(1-D)Y}{1-e(X)}\right] \right)$$

$$\widehat{ATE}_{IPW} = \frac{1}{n} \sum_{i=1}^n \left[\frac{D_i}{\hat{e}_i} - \frac{1-D_i}{1-\hat{e}_i} \right] Y_i$$

$$\widehat{ATT}_{IPW} = \frac{1}{n_T} \sum_{i=1}^n \left[D_i - (1-D_i) \frac{\hat{e}_i}{1-\hat{e}_i} \right] Y_i$$

- Both are **Horvitz–Thompson (1952)** forms.

Why *that* weight for the ATE?

We prove $E\left[\frac{D \cdot Y}{e(X)}\right] = E[Y^1]$. The $E[Y^0]$ half is symmetric.

$$E\left[\frac{D \cdot Y}{e(X)}\right] = E\left[\frac{D \cdot Y^1}{e(X)}\right]$$

switching equation (consistency): $D \cdot Y = D \cdot Y^1$

$$= E_X\left[E\left(\frac{D \cdot Y^1}{e(X)} \mid X\right)\right]$$

law of total expectation

$$= E_X\left[\frac{1}{e(X)} \cdot E(D \cdot Y^1 \mid X)\right]$$

$e(X)$ is fixed given X

$$= E_X\left[\frac{1}{e(X)} \cdot E(D \mid X) \cdot E(Y^1 \mid X)\right]$$

unconfoundedness (exchangeability): $D \perp\!\!\!\perp Y^1 \mid X$

$$= E_X\left[\frac{1}{e(X)} \cdot e(X) \cdot E(Y^1 \mid X)\right]$$

by definition, $E(D \mid X) = e(X)$

$$= E_X[E(Y^1 \mid X)] = E[Y^1]$$

cancel; law of total expectation

Why *that* weight for the ATT?

$$E[Y^1 \mid D=1] = E[Y \mid D=1] = \frac{E[DY]}{E[D]}.$$

$$E[Y^0 \mid D=1] = E_X \left[E(Y^0 \mid X, D=1) \mid D=1 \right]$$

law of total expectation, over $f(x \mid D=1)$

$$= E_X \left[E(Y^0 \mid X, D=0) \mid D=1 \right]$$

unconfoundedness — swap the arm

$$= E_X \left[E(Y^0 \mid X, D=0) \cdot \frac{e(X)}{\pi} \right]$$

Bayes: $f(x \mid D=1) = e(x)f(x)/\pi$

$$= E_X \left[\frac{E[(1-D)Y \mid X]}{1 - e(X)} \cdot \frac{e(X)}{\pi} \right]$$

$$E[(1-D)Y \mid X] = (1 - e(X))E[Y^0 \mid X, D=0]$$

$$= E \left[\frac{(1-D)Y e(X)}{(1 - e(X)) \pi} \right]$$

law of total expectation

Line 3 converts an average over the *treated* covariate distribution into one over the *whole* population.

Ten banks, split by size

Lecture 1's capital injections, now with a covariate: bank **size**.

Bank	Size	D	Y^0	Y^1	τ	observed Y
1	Small	1	10	12	2	12
2	Small	0	8	10	2	8
3	Small	0	9	11	2	9
4	Small	0	11	13	2	11
5	Small	0	12	14	2	12
6	Large	1	18	24	6	24
7	Large	1	19	25	6	25
8	Large	1	21	27	6	27
9	Large	1	22	28	6	28
10	Large	0	20	26	6	20

- Small banks are **rarely** rescued, 1 of 5; large banks **usually** are, 4 of 5. So $e(\text{Small}) = 0.2$ and $e(\text{Large}) = 0.8$.
- Small banks gain 2, large banks gain 6. Truth: $ATE = 4$, $ATT = 5.2$.
- Only the **bold** column is observed. Comparing rescued with not gives **11.2** — nearly three times the ATE .

The ATE rebuilds the population

Cell	N	e	injected	$\times 1/e$	controls	$\times 1/(1 - e)$
Small	5	0.2	1	$\times 5 = \mathbf{5}$	4	$\times 1.25 = \mathbf{5}$
Large	5	0.8	4	$\times 1.25 = \mathbf{5}$	1	$\times 5 = \mathbf{5}$
Total	10		5	10	5	10

- Bank 1 is the **only** small bank rescued, so it must stand in for all five. Its weight is $1/0.2 = 5$.
- Every weighted entry comes back to 5: each arm is blown up to the **full population of ten**. That is the *pseudo-population*, and in it $D \perp\!\!\!\perp X$ by construction.
- Before weighting, small banks are 20% of the injected sample and 80% of controls. After weighting, 50% of both — the imbalance causing the bias is gone.

The *ATT* reweights controls only

The *ATT* wants the **treated** distribution. The four large and one small rescued banks are already a sample from it, so they keep weight 1. Only the controls move.

Cell	controls	$\times e/(1 - e)$	=	injected
Small	4	0.25	1	1
Large	1	4.00	4	4

- Multiplying the controls by $e/(1 - e)$ makes their count match the injected count **exactly** — last column.
- Bank 10 is the lone large control and must cover **four** rescued banks, so it carries weight 4. Push e toward 1 and nobody is left to carry it: **this is where the *ATT* fails**, and why it needs only $e(X) < 1$ while the *ATE* needs $0 < e(X) < 1$.

Does it work?

$$\widehat{E}[Y^1] = \frac{1}{10} \left[\frac{12}{0.2} + \frac{24+25+27+28}{0.8} \right] = \frac{60+130}{10} = 19$$

$$\widehat{E}[Y^0] = \frac{1}{10} \left[\frac{8+9+11+12}{0.8} + \frac{20}{0.2} \right] = \frac{50+100}{10} = 15$$

$$\widehat{ATT} = \underbrace{\frac{12+24+25+27+28}{5}}_{\text{injected mean} = 23.2} - \underbrace{\frac{(8+9+11+12)(0.25)+20(4)}{4(0.25)+1(4)}}_{\text{controls} \times \text{odds} = 18.0}$$

- Truth: $ATE = 4$, $ATT = 5.2$. The naive SDO is $23.2 - 12 = \mathbf{11.2}$.
- IPW returns $19 - 15 = \mathbf{4}$ and $23.2 - 18.0 = \mathbf{5.2}$. Both exactly right.
- $ATT > ATE$ because large banks gain more, and large banks are where the injections went. **Selection on gains — the two estimands correctly disagree.**

Two ways to divide

Both estimators use the same weights $1/\hat{e}_i$. They differ **only in the denominator**.

$$\underbrace{\widehat{E}[Y^1]_{\text{HT}} = \frac{1}{N} \sum_i \frac{D_i Y_i}{\hat{e}_i}}_{\text{divide by } N}$$

$$\underbrace{\widehat{E}[Y^1]_{\text{Hájek}} = \frac{1}{\sum_i D_i / \hat{e}_i} \sum_i \frac{D_i Y_i}{\hat{e}_i}}_{\text{divide by the weights you got}}$$

- In the **population** they agree: $E[D/e(X)] = 1$, so the weights sum to N .
- In a **sample** they do not. $\sum_i D_i / \hat{e}_i$ is only an *estimate* of N , and it misses.
- Hájek's weights sum to one by construction, so it is always a genuine **weighted average of outcomes you actually observed**. HT carries no such guarantee.
- The default in Stata's `teffects ipw` and R's `WeightIt`.

Example 1

Four rescued banks. **Every one has lending growth** $Y = 10$, so the answer must be 10 for $\widehat{E[Y^1]}$. Here $N = 8$.

Bank	$\hat{e}$	weight $1/\hat{e}$	Y
A	0.50	2	10
B	0.50	2	10
C	0.50	2	10
D	0.25	4	10
sum		10	

$$\text{HT} = \frac{20 + 20 + 20 + 40}{8} = \frac{100}{8} = \mathbf{12.5}, \quad \text{Hájek} = \frac{100}{10} = \mathbf{10}.$$

- The weights sum to 10, not 8. HT divides by 8 anyway, returning $10 \times \frac{10}{8}$ — a number that appears nowhere in the data.

Example 2

Bank	$\hat{e}$	weight	share of weight	Y
A	0.50	2	3.6%	10
B	0.50	2	3.6%	10
C	0.50	2	3.6%	10
D	0.02	50	89.3%	24
sum		56	100%	

$$\text{HT} = \frac{1260}{8} = \mathbf{157.5}, \quad \text{Hájek} = \frac{1260}{56} = \mathbf{22.5}.$$

- D carries **89%** of the weight.
- **Normalization is necessary, not sufficient.** Check overlap separately: trim, or retreat to the *ATT* when the trouble is at $\hat{e} \approx 0$.

Propensity score matching

- Match each treated unit to controls with a **similar score**, then difference.
- **Caliper** matching: refuse any match farther than some tolerance (say 0.001 on the score), *discarding* the treated unit if none qualifies. **Radius** matching: average all controls inside the tolerance.
- Two problems with both:
 - ▶ The tolerance is **chosen by the researcher** (Huntington-Klein et al. 2021).
 - ▶ Dropping treated units **changes the estimand**: the *ATT* summarised the *original* treated sample.
- King and Nielsen (2019): two units can share a score, $e(X_i) = e(X_j)$, with opposite covariate profiles $X_i \neq X_j$. Matching on the scalar score can *increase* local imbalance relative to multivariate distance matching.

What the score does and does not buy

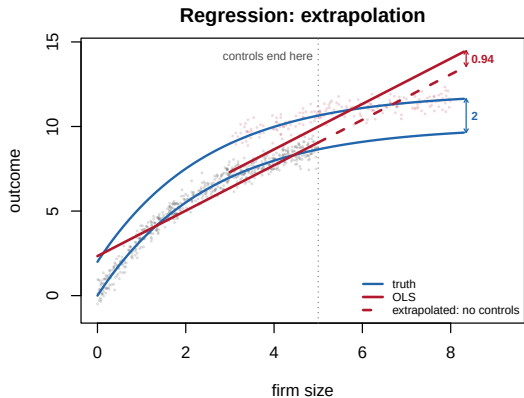
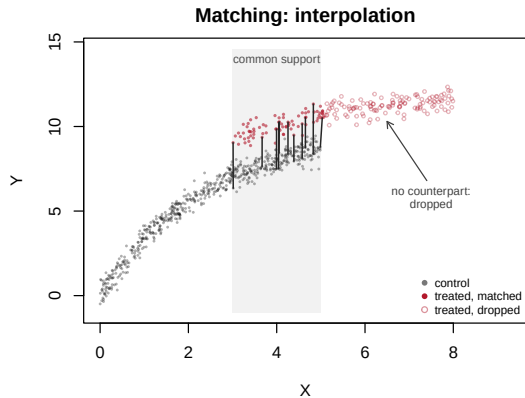
- It buys **dimension reduction**, not identification. If X omits a confounder, $e(X)$ omits it too.
- Regression hides missing common support behind linear extrapolation; plotting $\hat{e}(X)$ exposes it.
- A high pseudo- R^2 means covariates nearly predict treatment ($e(X) \rightarrow 0$ or 1): positivity breaks and IPW variance explodes. A low one means treatment is nearly independent of X — close to an RCT.
- IPW shifts the modelling assumption from the outcome equation $E[Y \mid D, X]$ to the assignment equation $P(D=1 \mid X)$.

2. Regression

Crossing to the other side of the map

- Everything so far has been an **interpolation** method: subclassification, matching, the propensity score. Each refuses to compare a treated unit with no counterpart; common support is the binding constraint.
- Regression sits on the other side: it needs no common support, so **the curse of dimensionality does not arise**.
- It buys that with a **functional form**. Where the data are silent the model speaks anyway: OLS imputes the counterfactual by **extrapolating** into regions with no comparable units.
- The trade: matching needs overlap; regression needs a correctly specified outcome model.
- Both still need unconfoundedness. Nothing in this section relaxes it.

Matching vs. regression



The bivariate case is easy; the multivariate case looks hard

- With one regressor,

$$Y_i = \beta_0 + \beta_1 X_i + u_i, \quad \hat{\beta}_1 = \frac{\text{Cov}(Y_i, X_i)}{\text{Var}(X_i)}.$$

- The coefficient is the slope of the best-fit line: a **covariance scaled by a variance**.
- Now add covariates:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \cdots + \beta_k X_{ki} + \varepsilon_i.$$

- Multivariate coefficients are **not** simple scaled covariances. But Frisch–Waugh–Lovell (Frisch and Waugh 1933; Lovell 1963) says they are scaled covariances of something else.

Frisch–Waugh–Lovell in three steps

Step 1. Auxiliary regression:

$$X_{1i} = \gamma_0 + \gamma_2 X_{2i} + \cdots + \gamma_K X_{Ki} + f_i.$$

$\hat{X}_{1i}$ is the part of X_{1i} the other covariates explain.

Step 2. Residualization:

$$\tilde{X}_{1i} = X_{1i} - \hat{X}_{1i}.$$

The variation in X_{1i} that is *uncorrelated* with everything else — **partialling out**.

Step 3. Regress the outcome on the residual.

$$Y_i = \alpha + \beta_1 \tilde{X}_{1i} + \varepsilon_i, \quad \hat{\beta}_1 = \frac{\text{Cov}(Y_i, \tilde{X}_{1i})}{\text{Var}(\tilde{X}_{1i})}.$$

- $\hat{\beta}_1$ is **identical** to the coefficient on X_{1i} in the full regression.

FWL is an identity, not an approximation

Simulation: 4,000 workers, earnings on a treatment indicator d , age, and high-school GPA, with a true effect of \$1,500.

	(1) Multivariate OLS	(2) FWL
d	1503.201 (8.249)	1503.201 (8.287)
age	1.173 (2.023)	—
gpa	9.081 (4.615)	—
N	4000	4000

- The estimate sits 0.39 above the true 1500 — sampling noise.
- The standard errors do *not* agree, and that is not sampling noise.

Residualize both sides

- Another version residualizes the outcome too:

$$\tilde{Y}_i = Y_i - \hat{Y}_i \quad \text{from} \quad Y_i = \lambda_0 + \lambda_2 X_{2i} + \cdots + \lambda_K X_{Ki} + v_i, \quad (1)$$

$$\text{then} \quad \tilde{Y}_i = \beta_1 \tilde{X}_{1i} + \varepsilon_i. \quad (2)$$

- Both versions return **exactly the same** $\hat{\beta}_1$.
- The double-residual version buys one more thing: **its residuals are those of the full regression.**

Three regressions, one coefficient

	$\hat{\beta}_d$	s.e.
<i>original</i> : the controls explain 0.1% of earnings		
full OLS	1503.201	8.249
single, $Y \sim \tilde{D}$	1503.201	8.287
double, $\tilde{Y} \sim \tilde{D}$	1503.201	8.247
<i>expanded</i> : the controls explain 93%		
full OLS	1503.201	8.249
single, $Y \sim \tilde{D}$	1503.201	91.493
double, $\tilde{Y} \sim \tilde{D}$	1503.201	8.247

- Residualize only D : the residuals still carry everything age and GPA explain.

Why FWL is worth knowing

- **It tells you what variation identifies your coefficient.** $\hat{\beta}_1$ uses only the part of X_1 that the controls cannot explain.
- **It makes the estimate visualizable.** Residualize *both* sides and scatter $\tilde{Y}$ against $\tilde{X}_1$: a multivariate coefficient becomes a picture whose slope is $\hat{\beta}_1$.¹
- **It is the engine behind much else.** Fixed effects, double machine learning and the weighting results below are all FWL in disguise.
- Angrist and Pischke (2009) call this **regression anatomy**, and it is the single most useful algebraic fact about OLS.

¹Filoso (2013) gives the worked proof and a Stata implementation.

Exogeneity, and the canonical OLS model

- Write the **canonical OLS model**:

$$Y_i = \alpha + \delta D_i + \gamma X_i + \varepsilon_i.$$

- For $\hat{\delta}$ to be causal, D_i must be *exogenous* $\Rightarrow$ Imbens and Rubin (2015) put it plainly: exogeneity “combines unconfoundedness with functional form and constant treatment effect assumptions that are quite strong, and arguably unnecessary.”
- Three assumptions, not one. You have met unconfoundedness; the other two arrived silently, in the act of writing the equation.

Taking conditional expectations

- Take expectations separately for the treated and control groups:

$$E[Y_i \mid D_i = 1, X_i] = \alpha + \delta E[D_i \mid D_i = 1, X_i] + \gamma_1 E[X_i \mid D_i = 1, X_i]$$

$$E[Y_i \mid D_i = 0, X_i] = \alpha + \delta E[D_i \mid D_i = 0, X_i] + \gamma_1 E[X_i \mid D_i = 0, X_i]$$

- Now replace Y with Y^1 or Y^0 as appropriate — and do the same for X .
- Why would X have potential outcomes? **If $X^1 \neq X^0$, then D causes X , and X is an outcome, not a covariate.** That is the bad-control problem.
- And allow the covariate to matter differently in the two worlds, γ_{11} against γ_{01} :

$$E[Y_i^1 \mid D_i = 1, X_i] = \alpha + \delta E[D_i \mid D_i = 1, X_i^1] + \gamma_{11} E[X_i^1 \mid D_i = 1, X_i^1]$$

$$E[Y_i^0 \mid D_i = 0, X_i] = \alpha + \delta E[D_i \mid D_i = 0, X_i^0] + \gamma_{01} E[X_i^0 \mid D_i = 0, X_i^0]$$

What $\gamma_{11} \neq \gamma_{01}$ actually means

- Let Y^1 be earnings with a college degree, Y^0 earnings without one, and X be age.
- γ_{11} is the age-earnings profile for **college graduates**; γ_{01} is that profile for the same people **had they not gone**.
- $\gamma_{11} \neq \gamma_{01}$ **is** heterogeneous treatment effects — the return to treatment varying with X .
- In finance this is unremarkable: the effect of index inclusion, a rating change or a governance shock varies with firm size. If it does, $\gamma_{11} \neq \gamma_{01}$.

Heterogeneous treatment effect bias

- OLS on the canonical model is a difference in conditional means:

$$\hat{\delta}_{OLS} = E[Y_i^1 \mid D_i = 1, X_i^1] - E[Y_i^0 \mid D_i = 0, X_i^0].$$

- Substitute the two expressions from the previous slide.

$$\hat{\delta}_{OLS} = \delta + \underbrace{\left(\gamma_{11}E[X_i^1 \mid D_i = 1, X_i^1] - \gamma_{01}E[X_i^0 \mid D_i = 0, X_i^0] \right)}_{\text{heterogeneous treatment effect bias}}$$

- This is what the canonical OLS model identifies. Not δ . δ plus a term that has nothing to do with omitted variables.
- Notice what is *not* in the bias term: any unobserved confounder. We assumed unconfoundedness holds perfectly and there is nothing left to omit.

The two assumptions you signed up for

The bias term vanishes if and only if **both** of the following hold.

1. **Covariates are not outcomes.** $X^1 = X^0 = X$: the covariates do not respond to treatment.
 2. **Treatment effects are homogeneous in X .** $\gamma_{11} = \gamma_{01} = \gamma_1$: the return to treatment does not vary with the covariates.
- Under A1 and A2, $\hat{\delta}_{OLS} = \delta$. Fail either one and the canonical model is biased.
 - **A1 has no repair.** Do not control for a collider or a mediator $\Rightarrow$ Lecture 2!
 - **A2 does have a repair**, and it is a change to the specification, not to the data. That is the next slide.
 - A2 is strong. Neither matching nor the propensity score required it: they average heterogeneous effects rather than assuming them away.

Regression adjustment: Oaxaca's answer

- The problem was never OLS. It was **one specification**: additive and separable.
- The **Kitagawa–Oaxaca–Blinder decomposition** splits a mean gap into differences in *endowments* (education, experience) and differences in the *returns* to them.
- Słoczyński (2015, 2020) showed the decomposition is also a **treatment effects estimator**, identifying both the *ATE* and the *ATT*. That is what “regression adjustment” means.

Deriving the specification

- Write both potential outcomes, letting X act differently on each:

$$Y_i^0 = \alpha + \gamma_{01}X_i + \varepsilon_i$$

$$Y_i^1 = \alpha + \gamma_{01}X_i + \delta D_i + \gamma_{11}X_i \cdot D_i + \varepsilon_i$$

- Take conditional expectations, then form $\delta = E[Y_i^1 \mid D_i, X_i] - E[Y_i^0 \mid D_i, X_i]$:
$$\begin{aligned} &= (\alpha + \gamma_{01}E[X_i \mid D_i=0, X_i] + \delta + \gamma_{11}E[X_i \cdot D_i \mid D_i=1, X_i]) - (\alpha + \gamma_{01}E[X_i \mid D_i=0, X_i]) \\ &= \delta + \gamma_{11}E[X_i \cdot D_i \mid D_i = 1] \end{aligned}$$
- The γ_{01} terms cancel because the *same* coefficient appears in both potential outcomes. What remains is the interaction — exactly the term the canonical model omitted.

The regression adjustment estimator

To identify the *ATE* and the *ATT* under unconfoundedness **with heterogeneous effects**, run

$$Y_i = \alpha + \delta D_i + \gamma_{01} X_i + \gamma_{11} D_i \cdot X_i + \varepsilon_i.$$

- Interact the treatment with **every** covariate. Then

$$\widehat{ATE} = \hat{\delta}, \quad \widehat{ATT} = \hat{\delta} + \hat{\gamma}_{11} E[X_i \mid D_i = 1],$$

where $E[X_i \mid D_i = 1]$ is the covariate mean **in the treated group only**.²

- The difference between the two estimands is now visible and mechanical: the same coefficients, evaluated at a different covariate distribution.

²The *ATE* result requires X to be demeaned in the full sample. If you interact with terms that are not mean-zero — squares, cross-products — add them back: $\widehat{ATE} = \hat{\delta} + \sum_k \hat{\gamma}_{1k} \bar{X}_k$.

Does it work?

Simulation: 5,000 workers, 5,000 with a college degree; two exogenous covariates, age and GPA, with **different means in the treated and control groups**; both recentred, plus their squares and their interaction. Then

$$Y^0 = 15000 + 10.25 \text{ age} - 10.5 \text{ age}^2 + 1000 \text{ gpa} - 10.5 \text{ gpa}^2 + 500 (\text{age} \cdot \text{gpa}) + e$$

$$Y^1 = Y^0 + 2500 + 100 \text{ age} + 1100 \text{ gpa}$$

- Because age and GPA are recentred, $E[Y^1 - Y^0] = \$2,500$ exactly. The *ATT* is **not** \$2,500: the treated group differs in age and GPA composition, and the effect varies with both.
- The canonical model here controls for age, age², GPA, GPA² and their interaction. Every covariate needed for unconfoundedness is in it. Nothing is omitted.

When is the canonical model actually wrong?

Simulation, varying the **share treated**:

Share treated	0.20	0.35	0.50	0.65	0.80
True <i>ATE</i>	2500	2500	2500	2500	2500
True <i>ATT</i>	1813	1942	2070	2199	2328
Canonical OLS	2097	2310	2500	2689	2902
Regression adj. (<i>ATE</i>)	2500	2500	2499	2499	2499
Regression adj. (<i>ATT</i>)	1812	1941	2069	2198	2327

- Regression adjustment recovers **both** estimands at every split.
- At exactly 50/50 the canonical model lands on the *ATE* — **by coincidence of the design, not by validity.**

What regression adjustment does not buy

- Regression adjustment removes the **constant-effects** assumption. It does not remove the **functional form** assumption, and cannot.
- Inside the model is a model of the missing potential outcome. The true Y^0 contained age, age², GPA, GPA² and age · GPA — so the treatment had to be interacted with **all five**.
- Drop one interaction and it breaks. With the effect also varying in age · GPA, the fully interacted model returns 2576 against a true *ATE* of 2575; omitting the single $D \cdot (\text{age} \cdot \text{gpa})$ term returns 2368.
- **The functional-form requirement replaces the common-support requirement. It is not free.** You have swapped a checkable condition for an uncheckable one.
- This is the honest summary of the trade made at the start of the section.

Three things wrong with the canonical model

1. **It assumes constant treatment effects.** Neither matching nor the propensity score asked for this. Regression adjustment removes it, at the cost of interacting treatment with everything.
2. **Its weights are wrong.** Słoczyński (2022) shows the coefficient is $w_1 \cdot ATT + w_0 \cdot ATU$, with each group's weight **inversely** proportional to its size; **it is generally none of the three.**
3. **It delivers only one number.** One treatment indicator, one coefficient. Regression adjustment delivers the *ATE* and the *ATT* from the same fit.

3. What Matching Cannot Do

Matching creates no new variation

- Matching **does not introduce exogenous variation**. It reorganizes and reweights the data you already have, using observables you already measured.
- Consequence: if your OLS regression was biased by endogeneity, **the matched estimate is biased too**.
- Contrast this with the designs still to come:
 - ▶ **IV** imports variation from outside the model.
 - ▶ **RDD** exploits assignment that is locally as good as random.
 - ▶ **DiD** differences out fixed unobservables.
- Matching belongs to a different category. It is a *conditioning strategy*, like regression — not a research design.

What it cannot fix

Matching is powerless against:

- **Unobserved confounders.** You cannot match on what you cannot measure.
- **Reverse causality.** Matching on X says nothing about whether Y caused D .
- **Measurement error** in treatment or outcome.
- **Bad controls.** Matching on a post-treatment variable is exactly as damaging as controlling for it — Lecture 2 applies unchanged, and the collider F of Lecture 3a §2 is the warning.

You cannot match on what you cannot measure.

So what is it good for?

Given all that, matching still earns its place:

1. **A non-parametric benchmark.** It estimates the effect without assuming a functional form. If matching and OLS disagree sharply, your specification is doing more work than you thought.
 2. **Common-support discipline.** Matching refuses to compare units with no counterpart. Regression silently extrapolates into regions with no data.
 3. **Diagnostic value.** The propensity score distribution reveals overlap problems that a regression table hides completely.
 4. **Robustness when the functional form is wrong.** Matching can outperform a misspecified regression in finite samples.
- Use it as a *complement* to regression and a check on it — not as an identification strategy.

4. The NSW Demonstration

The 1970s empirical crisis

- The methods in this lecture were not adopted because they were elegant. They were adopted after a crisis of confidence in observational work.
- Job-training programs in the 1970s were evaluated observationally, and the estimates were all over the place — depending, apparently, on the analyst.
- The **National Supported Work (NSW) Demonstration** was different: it *randomized* participation. So the experimental effect was known.
- That created a **benchmark**: could observational methods, applied to the same treated units, recover the experimental answer?
- Everything in this lecture gets graded against it.

What LaLonde did

- LaLonde (1986) ran the decisive test: take the NSW experiment, **throw away the experimental control group**, and replace it with comparison groups built from survey data (CPS and PSID).
- He applied the standard observational toolkit and compared the results to the experimental benchmark of roughly **\$1,800** in annual earnings gains.
- The estimates ranged from a fraction of the benchmark to **the wrong sign**, depending on how the comparison group was built.
- The paper was devastating, and a major reason the profession moved toward IV, RDD and DiD.

Dehejia and Wahba: not so fast

- **Dehejia and Wahba (1999, 2002)** revisited LaLonde's data with propensity score methods.
- Their finding: the observational estimates *did* recover the experimental benchmark reasonably well — **once common support was imposed**.
- Much of LaLonde's failure was an **overlap** problem, not only confounding: the CPS and PSID groups held huge numbers of people utterly unlike NSW participants.
- Discard the non-comparable units and the methods perform far better.
- This is why the Lecture 3a §3 distinction matters: two assumptions were failing, and only one is checkable.

Smith and Todd: not so fast either

- **Smith and Todd (2005)** re-examined the Dehejia–Wahba result and found it fragile.
- The success was **sensitive to the subsample** analysed and to the **propensity score specification** chosen. Reasonable alternative choices undid it.
- Their charge is uncomfortable and worth stating plainly: the specification that worked may have been found *because* the experimental answer was already known.
- You cannot run that search in real research: there is no benchmark to check against — that is the point of doing the study.

The honest summary

- Selection on observables **can** work — when overlap is genuine and the confounders are truly measured.
- Neither condition is verifiable in the setting where you need them:
 - ▶ Unconfoundedness is **untestable**, always.
 - ▶ Common support is **checkable**.
- So the practical lesson is asymmetric: do the check you can do, and argue explicitly for the assumption you cannot.
- **A useful exercise:** the NSW data are freely available. Apply every method here and compare to the \$1,800 benchmark — the fastest way to build instincts about when these methods hold up.

Wrapping Up

Summary [Part 1]: the score and what it does

- **The propensity score** reduces X to one number and defeats the curse of dimensionality for *conditioning*. It does nothing for **identification** — unconfoundedness is still assumed, never delivered.
- **Judge the score by balance achieved, not by fit.** A high pseudo- R^2 with bad balance is a bad score; the reverse is fine. The stage is blind to Y , which is what makes the specification search legitimate.
- **Balance means the distribution, not the mean.** Standardized differences *and* variance ratios. Our Volatility covariate had $SMD = -0.010$ and a variance ratio of 2.27.
- **IPW** reweights into a pseudo-experiment: $1/\hat{e}$ and $1/(1 - \hat{e})$ build the population; $\hat{e}/(1 - \hat{e})$ on controls alone rebuilds the *treated* distribution. Watch the tails, and normalize.
- **Propensity score matching** is the one use with a serious case against it: King and Nielsen (2019) show it targets complete randomization when blocking was available, so pruning can *increase* imbalance.

Summary [Part 2]: the verdict

- **Regression is a conditioning strategy too**, not a rival family. FWL shows it uses only the **residual variation in D** , and Słoczyński's weights show the resulting coefficient is generally neither the *ATE* nor the *ATT*.
- **Matching creates no new exogenous variation**. If OLS was endogenous, matching is endogenous. It cannot fix an unobserved confounder, reverse causality, or a bad control.
- **NSW**: these methods can work when overlap is real and the confounders are measured — and you can only ever verify one of those two.
- Next lecture: designs that *create* variation instead of assuming it away.

A checklist for your own paper

1. State the **estimand** before the estimator. *ATE*, *ATT*, or *ATU* — and say why the policy question needs that one.
2. Defend the **assignment mechanism** in institutional terms. Name the confounders and say how you measured them.
3. Justify every covariate as **pre-treatment**. Any post-treatment control needs an argument, not a footnote.
4. **Plot the overlap**. Report what you trimmed and how the estimand changed.
5. Report **balance**, not fit: standardized differences and variance ratios, before and after.
6. Report **matching and regression**. If they disagree, diagnose whether it is weights or extrapolation.
7. Run a **sensitivity analysis**, and calibrate the threshold against an observed covariate.
8. Run a **placebo** on a pre-treatment period.
9. Say plainly what would break the design — and what unobservable would have to exist.

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